Epiphytes

The plants which are found growing on other plants without harming them are called epiphytes. They are commonly found in tropical rain forest.

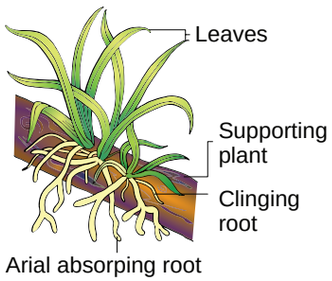


Figure 6.11: An epiphytic plant-*Vanda*

The epiphytic higher plant (**Orchid**) gets its nutrients and water from the atmosphere with the help of the hygroscopic roots which contain special type of spongy tissue called **Velamen.** It prepares its own food and does not depend on the host. Using the host plant only they support and does not harm it in any way.

- Many orchids, ferns, lianas, hanging mosses, *Peperomia*, money plant and *Usnea* (Lichen) are some of the examples of epiphytes.
- Spanish Moss –*Tillandsia* grows on the bark of Oak and Pine trees.



Proto Cooperation

An interaction between organisms of different species in which both organisms benefit but neither is dependent on the relationship. Example: Soil bacteria / fungi and plants growing in the soil.

Negative interactions

When one of the interacting species is benefitted and the other is harmed, it is called **negative interaction** . Examples: predation, parasitism, competition and amensalism.

a. Predation: It is an interaction between two species, one of which captures, kills and eats up the other. The species which kills is called a **predator** and the species which is killed is called a **prey**. The predator is benefitted while the prey is harmed.

Examples:

- A number of plants like *Drosera* (Sun dew Plant), *Nepenthes* (Pitcher Plant), *Dionaea* (Venus fly trap), *Utricularia* (Bladder wort) and *Sarracenia* are predators which consume insects and other small animals for their food as a source of nitrogen. They are also called as **insectivorous plants**.

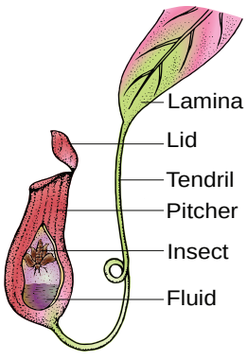


Figure 6.12: Pitcher plant – with insect

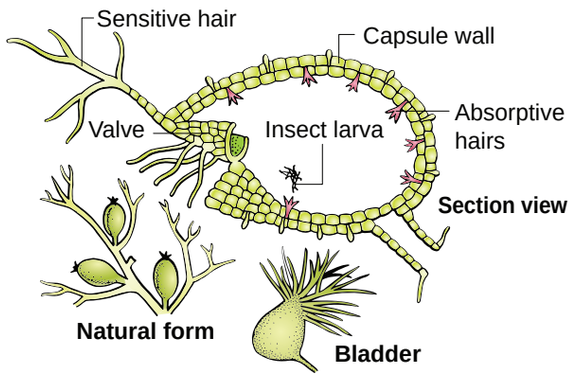


Figure 6.13: Insectivorous plant *Utricularia*

- Many herbivores are **predators**. Cattles, Camels, Goats etc., frequently browse on the tender shoots of herbs, shrubs and trees. Generally annuals suffer more than the perennials. Grazing and browsing may cause remarkable changes in vegetation. Nearly 25 percent of all insects are known as phytophagous(feeds on plant sap and other parts of plant)
- Many **defense mechanisms** are evolved to avoid their predations by plants. Examples: *Calotropis* produces highly poisonous cardiac glycosides, Tobacco produces nicotine, coffee plants produce caffeine, *Cinchona* plant produces quinine. Thorns of *Bougainvillea*, spines of *Opuntia*, and latex of cacti also protect them from predators.

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